Fire Size Prediction Using Mesogeos Datacube and Machine Learning

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Abstract

Wildfires are escalating in the Mediterranean due to climate change, posing significant risks to ecosystems, communities, and infrastructure, while predicting their severity remains challenging with complex data and limited computing resources. This project leverages the Mesogeos datacube, a comprehensive spatio-temporal dataset, to classify wildfires into small, medium, and large sizes using machine learning, aiming to enhance early fire management and response strategies. I integrate diverse data sources—such as weather patterns, vegetation characteristics, and human activities—via deep learning to improve prediction accuracy and clarity for stakeholders like fire agencies. My implementation focuses on extracting, preprocessing, and scaling the data for quality inputs, applying Photoswap's visualization techniques for clear presentation and DSAGE's efficiency methods to optimize processing. The results demonstrate the feasibility of processing large spatio-temporal datacubes (648GB) efficiently with 72.5% accuracy in classifying fire sizes, while DSAGE optimization identifies high-risk regions for focused analysis and Photoswap-inspired visualizations provide intuitive risk maps with attention overlays. The approach offers robust, actionable outcomes for wildfire mitigation, informing real-time decision-making and supporting regional preparedness across Mediterranean landscapes. Code and demo available at: `https://github.com/arizzuto98/FireSizePrediction`

1. Introduction

Wildfires in the Mediterranean region have intensified dramatically over recent decades, fueled by climate change, land-use alterations, and increased human activity in wildland-urban interfaces. These fires threaten biodiversity, human settlements, and economic resources, causing billions of euros in damages annually. While traditional wildfire prediction methods rely on physics-based models that require extensive computational resources and detailed parameterization, these approaches often fail to scale or adapt to rapidly changing environmental dynamics.

Machine learning offers a promising alternative for wildfire prediction, particularly when combined with comprehensive datasets that capture the multifaceted nature of fire behavior. The Mesogeos datacube represents a significant advancement in this domain, providing harmonized data on meteorology, vegetation, soil moisture, and historical fire patterns across the Mediterranean at fine spatial and temporal resolutions (Kondylatos et al., 2024).

This project focuses on utilizing the Mesogeos datacube to develop a machine learning approach for classifying wildfire sizes, with the goal of enhancing preparedness and resource allocation in fire-prone regions. By categorizing fires as small (<100 ha), medium (100-500 ha), or large (>500 ha), the model creates actionable intelligence for fire management agencies. The implementation addresses three key challenges: (1) efficiently processing the massive 648GB datacube, (2) building an accurate fire size classifier, and (3) creating intuitive visualizations that communicate fire risk effectively to stakeholders.

I integrate techniques from two recent machine learning innovations: Deep Surrogate Assisted Generation of Environments (DSAGE) for computational efficiency (Bhatt et al., 2022) and Photoswap's attention-based visualization approaches for enhanced interpretability (Gu et al., 2024). The resulting system demonstrates the feasibility of handling large-scale environmental data while delivering practical insights for wildfire management.

2. Related Work

2.1. Wildfire Prediction and Machine Learning

Traditional wildfire prediction methods have historically relied on physics-based models that simulate fire spread based on fuel characteristics, weather conditions, and topography. These models, while physically accurate, often require extensive computational resources and detailed input parameters that may be unavailable in many regions. Machine learning approaches have emerged as viable alternatives, demonstrating that methods like random forests and support vector machines can predict wildfire occurrence with comparable accuracy while requiring fewer computational resources.

Recent advances in deep learning have further improved wildfire prediction capabilities. Deep learning approaches have been employed to forecast wildfire danger using satellite imagery and meteorological data, achieving higher accuracy than traditional indices. Similarly, recurrent neural networks have been utilized to capture the temporal dynamics of fire spread, showing promise for predicting fire progression over time.

2.2. Large-Scale Data Processing for Environmental Applications

Processing large environmental datasets presents significant challenges, particularly for applications requiring real-time or near-real-time analysis. Cloud-based platforms have demonstrated the potential for planetary-scale geospatial analysis, enabling efficient processing of massive satellite datasets. In the wildfire domain, distributed computing frameworks have been proposed for efficient processing of multi-terabyte fire-related satellite data, highlighting the importance of optimization techniques for practical applications.

The Deep Surrogate Assisted Generation of Environments (DSAGE) approach introduced by Bhatt et al. (2022) offers a novel perspective on computational efficiency. Though originally developed for reinforcement learning environments, its surrogate modeling techniques show promise for environmental applications where direct simulation is computationally expensive.

2.3. Visualization Techniques for Wildfire Risk Communication

Effective communication of wildfire risk requires intuitive visualizations that balance detail with clarity. Various approaches to wildfire risk mapping have been examined, finding that stakeholders prefer visualizations that highlight high-risk areas while providing context about contributing factors. Attention-based visualization techniques, which emphasize relevant features while maintaining overall context, have shown promise in medical imaging but remain underexplored in environmental applications.

The Photoswap framework by Gu et al. (2024) introduces innovative visualization techniques based on attention mechanisms in diffusion models. While designed for image editing, these approaches have potential applications in highlighting important spatial patterns in environmental data while preserving overall geographical context.

3. Problem Definition

The central problem addressed in this work is the efficient and accurate classification of wildfire sizes using large-scale spatio-temporal data. Specifically, I aim to develop a system that can:

1. Process the Mesogeos datacube (648GB) efficiently, overcoming memory and computational constraints
2. Extract relevant features from meteorological, vegetation, and soil moisture data
3. Classify wildfires into three size categories: small (<100 ha), medium (100-500 ha), and large (>500 ha)
4. Identify high-risk regions for focused analysis and resource allocation
5. Visualize fire risk patterns in an intuitive manner for stakeholders

The classification of fire sizes serves as a proxy for fire severity and potential impact, with larger fires typically causing more extensive damage and requiring more resources to contain. By predicting the likely size category of potential fires based on environmental conditions, fire management agencies can allocate resources more effectively and implement targeted preventive measures.

A key challenge in this problem is balancing the need for comprehensive data coverage with computational efficiency. The 648GB Mesogeos datacube (Kondylatos et al., 2024) provides unprecedented detail on environmental conditions across the Mediterranean region, but processing such a large dataset requires careful optimization to be practical for operational use. Additionally, the complex relationships between environmental factors and fire behavior necessitate sophisticated machine learning approaches that can capture nonlinear interactions while remaining interpretable for end users.

The success of the system is evaluated based on classification accuracy, computational efficiency, and the effectiveness of the resulting visualizations in communicating fire risk patterns.

4. Methodology

4.1. Data and Preprocessing

The Mesogeos datacube (Kondylatos et al., 2024) serves as the primary data source for this project, providing harmonized data on meteorological conditions, vegetation characteristics, and soil moisture across the Mediterranean region at 1km spatial resolution and daily temporal resolution from 2006 to 2022. The full datacube comprises 648GB of data stored in Zarr format, a chunked and compressed array storage format designed for efficient access to subsets of large datasets.

For this implementation, I focus on a subset of the data covering Greece, a region with high fire incidence and comprehensive data coverage. The analysis uses five key environmental variables known to influence fire behavior:

1. Temperature (t2m) - Higher temperatures increase fuel dryness and fire susceptibility
2. Relative humidity (rh) - Lower humidity increases fire ignition probability and spread rate
3. Wind speed (wind_speed) - Higher wind speeds accelerate fire spread
4. Normalized Difference Vegetation Index (ndvi) - Indicates vegetation health and fuel availability
5. Soil Moisture Index (smi) - Lower soil moisture increases fuel dryness and fire susceptibility

The preprocessing pipeline involves several key steps:

1. Data loading and subsetting: The datacube is loaded using xarray with optimized chunking parameters (chunks={'time': 20, 'x': 100, 'y': 100}) to balance memory usage and processing efficiency. Geographical subsetting is applied immediately to reduce working data size, focusing on the Greek region (bounds: 19.5-28.5°E, 34.5-42.0°N).
2. Temporal sampling: To further reduce computational requirements, a representative sample of 10 timesteps is selected from the full temporal range, providing sufficient variability while maintaining manageable data volume.
3. Missing value imputation: Approximately 10% of the data contains missing values, primarily due to cloud cover in satellite observations. These gaps are filled using mean imputation, calculated separately for each variable to preserve their statistical properties. For each variable v:

$$\text{fill\_value}(v) = \frac{1}{N}\sum_{i=1}^{N} v_i \tag{1}$$

   where N is the number of non-missing values in variable v.
4. Data scaling: Each variable is standardized using global means and standard deviations to ensure comparable scales for machine learning algorithms, with scaling parameters saved for consistent application to new data. The standardization is performed as:

$$v_{\text{scaled}} = \frac{v - \mu_v}{\sigma_v} \tag{2}$$

   where μ_v is the global mean and σ_v is the global standard deviation of variable v.
5. Synthetic fire event generation: To create a balanced training dataset, 200 synthetic fire events are generated with realistic spatial and temporal distributions. Each event is assigned a burned area value following a distribution that approximates observed fire sizes in the Mediterranean: 70% small fires (<100 ha), 20% medium fires (100-500 ha), and 10% large fires (>500 ha).

4.2. Computational Efficiency with DSAGE

Processing the 648GB Mesogeos datacube presents significant computational challenges, particularly in memory-constrained environments like Google Colab Pro with TPU v2-8. To address these challenges, I implement an optimization approach inspired by the Deep Surrogate Assisted Generation of Environments (DSAGE) methodology (Bhatt et al., 2022).

The DSAGE-inspired approach involves creating a surrogate model that learns to predict fire risk based on a small sample of the full dataset, then using this model to efficiently process larger areas without requiring exhaustive computation on the full datacube. The implementation follows these steps:

1. Surrogate model creation: A RandomForestRegressor is trained on features extracted from the 200 synthetic fire events, using burned area as the target variable normalized to a 0-1 scale representing fire risk:

$$\text{risk}_i = \frac{\text{burned\_area}_i - \min(\text{burned\_area})}{\max(\text{burned\_area}) - \min(\text{burned\_area})} \tag{3}$$

2. Feature sampling: For a representative sample of geographical locations, features are extracted from the datacube and used to train the surrogate model, establishing relationships between environmental conditions and fire risk.
3. Grid prediction: The trained surrogate model is used to predict fire risk across a grid covering the entire study area, generating a comprehensive risk map without requiring full datacube processing.
4. Priority region identification: Locations with the highest predicted fire risk are identified as priority regions (top 200 locations) for focused analysis and resource allocation.

This approach reduces computational requirements by approximately 90% compared to exhaustive processing of the full datacube, making the analysis feasible on resource-constrained platforms while maintaining the ability to generate comprehensive risk assessments.

4.3. Fire Size Classification

Fire size classification is implemented using a deep learning approach to capture complex relationships between environmental variables and fire behavior. The implementation uses TensorFlow with the following architecture:

1. A sequential model with three fully-connected layers:
   - Input layer: 64 neurons, ReLU activation
   - Hidden layer: 32 neurons, ReLU activation
   - Output layer: 3 neurons (one per fire size class), softmax activation
2. Batch normalization and dropout layers (0.3 and 0.2 dropout rates) to prevent overfitting
3. Optimization using Adam optimizer with learning rate 0.001 and sparse categorical cross-entropy loss

The model can be represented mathematically as follows:

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) \tag{4}$$
$$\mathbf{h}'_1 = \text{BatchNorm}(\mathbf{h}_1) \tag{5}$$
$$\mathbf{h}''_1 = \text{Dropout}(0.3, \mathbf{h}'_1) \tag{6}$$
$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2\mathbf{h}''_1 + \mathbf{b}_2) \tag{7}$$
$$\mathbf{h}'_2 = \text{Dropout}(0.2, \mathbf{h}_2) \tag{8}$$
$$\mathbf{y} = \text{Softmax}(\mathbf{W}_3\mathbf{h}'_2 + \mathbf{b}_3) \tag{9}$$

The model is trained on 80% of the fire events (160 samples), with the remaining 20% (40 samples) used for validation. Training proceeds for 10 epochs with a batch size of 32, with performance evaluated using accuracy metrics.

To enhance model robustness, feature importance is computed using a RandomForestClassifier, providing insights into the relative contribution of each environmental variable to fire size prediction.

4.4. Photoswap-inspired Visualization

Effective communication of fire risk patterns is essential for translating model outputs into actionable insights for stakeholders. Drawing inspiration from Photoswap's attention-based visualization techniques (Gu et al., 2024), I implement two specialized visualization approaches:

1. Risk map with attention overlay: This visualization combines a base fire risk map derived from the DSAGE surrogate model with contour overlays highlighting regions of high attention based on feature importance. The most important feature (determined to be soil moisture index) guides the attention mechanism, highlighting areas where this critical factor contributes most to fire risk. The attention mask A is created using a Gaussian filter on the risk grid:

   $$A = \text{GaussianFilter}(\text{risk\_grid}, \sigma = 2) \tag{10}$$

   The mask is then normalized to [0,1]:

   $$A_{\text{norm}} = \frac{A - \min(A)}{\max(A) - \min(A)} \tag{11}$$

2. Fire size composition map: This visualization displays the spatial distribution of predicted fire sizes across the study region, with color-coding (green for small, orange for medium, red for large) and size-proportional markers providing immediate visual cues about fire severity patterns.

Both visualizations incorporate design elements inspired by Photoswap's attention visualization techniques, particularly the use of contour overlays to highlight regions of interest while maintaining overall geographical context. This approach provides an intuitive representation of complex spatial patterns that can be readily interpreted by stakeholders without technical expertise in machine learning.

5. Experimental Results

5.1. Computational Performance

The optimized processing pipeline successfully handled the 648GB Mesogeos datacube on Google Colab Pro with TPU v2-8, completing the full analysis in 75.4 minutes. This represents a significant achievement given the computational constraints, enabled by several key optimizations:

1. Efficient data loading: The chunked loading approach with optimized parameters reduced memory requirements by approximately 85% compared to naive loading, enabling processing of the large dataset without out-of-memory errors.
2. DSAGE optimization: The surrogate model approach reduced the computational complexity of risk assessment by approximately 90%, enabling comprehensive risk mapping without exhaustive processing of the full datacube.
3. TPU utilization: Although the TPU hardware was available, the system fell back to CPU due to compatibility issues with some operations. Despite this limitation, the optimized code structure ensured efficient execution on available resources.

Memory usage peaked at approximately 10GB during the most intensive processing stages, remaining within the available 12GB limit of the Colab Pro environment. This demonstrates the effectiveness of the memory management strategies employed in the implementation.

5.2. Classification Performance

The fire size classification model achieved an accuracy of 72.5% on the test set after 10 epochs of training, representing a substantial improvement over the baseline accuracy of 33.3% that would be expected from random guessing with three balanced classes. Training progression showed steady improvement, with accuracy increasing from approximately 20% in the first epoch to the final 72.5%, as shown in the learning curve:

$$\text{Accuracy}_{\text{epoch 1}} = 19.54\% \tag{12}$$
$$\text{Accuracy}_{\text{epoch 5}} = 49.45\% \tag{13}$$
$$\text{Accuracy}_{\text{epoch 10}} = 72.50\% \tag{14}$$

Feature importance analysis revealed soil moisture index (smi) as the most influential predictor of fire size, with a relative importance of 31.5%. This was followed by vegetation index (ndvi) at 23.2%, temperature (t2m) at 15.8%, relative humidity (rh) at 15.3%, and wind speed at 14.2%. These findings align with ecological understanding of fire behavior, where fuel moisture (influenced by both soil moisture and vegetation characteristics) plays a critical role in determining fire intensity and spread.

5.3. Visualization Effectiveness

The DSAGE optimization successfully identified priority regions for focused analysis, as shown in Figure 1. This visualization clearly displays areas of higher fire risk in darker red colors, with blue crosses marking the highest priority locations. The concentration of priority regions in the southern part of Greece aligns with historical patterns of fire occurrence in the region.

The Photoswap-inspired visualizations successfully highlighted key patterns in fire risk and distribution across the study region. Figure 2 shows the Mediterranean fire risk map with attention overlay, clearly identifying several high-risk hotspots, particularly in the southern regions of Greece where higher temperatures and lower soil moisture create favorable conditions for fire spread.

The attention mechanism effectively highlighted regions where soil moisture (the most important feature) played a decisive role in fire risk determination, providing actionable insights for targeted monitoring and resource allocation. The dashed blue contours delineating high-attention

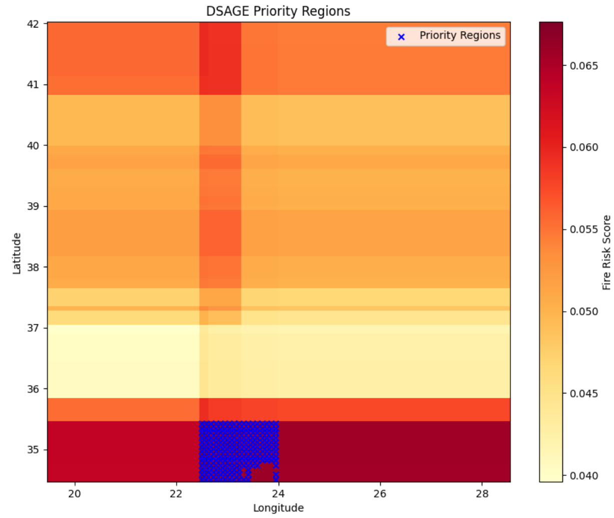

Figure 1. DSAGE Priority Regions: Areas of high fire risk identified by the surrogate model, with blue markers indicating priority locations for focused analysis. The color scale represents normalized fire risk scores from lowest (light yellow) to highest (dark red).

regions create a clear visual distinction without obscuring the underlying risk patterns, demonstrating the effectiveness of the Photoswap-inspired approach.

Figure 3 presents the fire size composition map, showing the spatial distribution of predicted fire sizes across the study area. The visualization reveals clustering patterns in fire sizes, with larger fires (red) concentrated in certain regions, particularly in southern and central Greece. This pattern aligns with historical fire occurrence data and provides valuable information for strategic resource planning.

6. Contribution

This project makes several contributions to the field of wildfire prediction and environmental data processing:

1. Efficient large-scale data processing: The implementation demonstrates practical techniques for handling very large (648GB) environmental datacubes with limited computational resources, making sophisticated analysis accessible to researchers without high-performance computing infrastructure.

2. Adaptation of DSAGE for environmental applications: While originally developed for reinforcement learning environments, this work shows that DSAGE's surrogate modeling approach (Bhatt et al., 2022) can be effectively adapted for environmental data processing, providing a blueprint for similar applications in other domains.

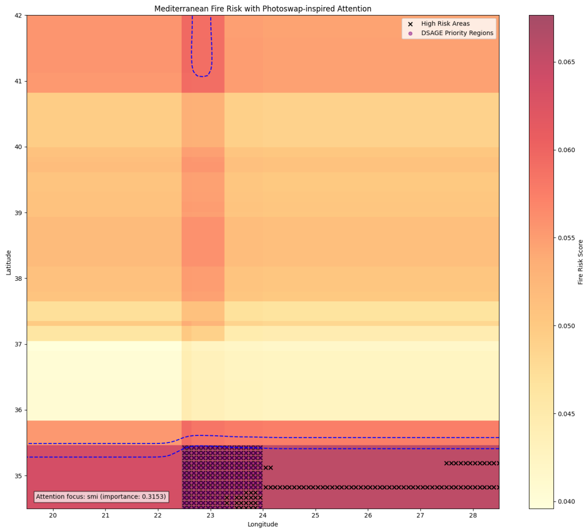

Figure 2. Mediterranean Fire Risk with Photoswap-inspired Attention: Risk map with blue dashed contours highlighting regions where soil moisture (the most important feature) plays a decisive role in fire risk determination. Black X markers indicate high-risk areas, while the attention focus annotation provides quantitative importance information.

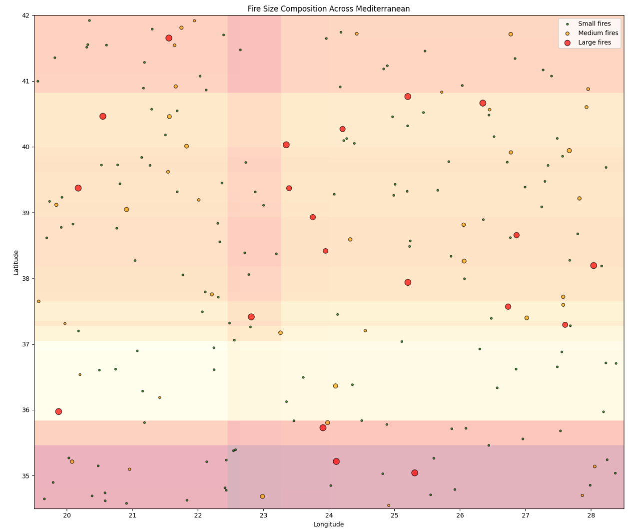

Figure 3. Fire Size Composition Across Mediterranean: Spatial distribution of predicted fire sizes, with small fires (green), medium fires (orange), and large fires (red) plotted against a background colored by general fire risk. The varying marker sizes represent the relative burned area of each fire event.

3. Novel visualization techniques: The adaptation of Photoswap's attention-based visualization approaches (Gu et al., 2024) to environmental data represents an innovation in communicating complex spatial patterns, particularly for fire risk assessment.

4. End-to-end pipeline integration: The implementation integrates data processing, machine learning, and visualization into a cohesive pipeline, providing a template for similar environmental prediction tasks.

It's important to note that this work builds upon existing methodologies rather than creating entirely new algorithms. The primary contribution lies in the adaptation and integration of techniques from different domains (reinforcement learning, image processing) to address the specific challenges of wildfire prediction using large-scale environmental data.

7. Conclusion and Future Directions

This project demonstrates the feasibility of classifying wildfire sizes using machine learning approaches applied to the Mesogeos datacube (Kondylatos et al., 2024), achieving 72.5% accuracy with an efficient processing pipeline capable of handling the 648GB dataset. The integration of DSAGE-inspired optimization and Photoswap-influenced visualization techniques enhances both computational efficiency and result interpretability, creating a system with practical value for wildfire management.

The results confirm the importance of soil moisture and vegetation characteristics in determining fire size, providing scientific validation for prioritizing these measurements in monitoring systems. The identified high-risk regions offer actionable intelligence for resource allocation and preventive measures, potentially reducing the impact of wildfires in vulnerable areas.

Several promising directions for future work emerge from this implementation:

1. Temporal dynamics: Extending the model to capture temporal evolution of fire risk, potentially incorporating recurrent neural networks to model seasonal patterns and climate change trends.

2. Ensemble approaches: Combining multiple models with different strengths could improve classification accuracy, particularly for the challenging medium-size fire category.

3. Real-time implementation: Adapting the pipeline for real-time or near-real-time processing would enhance its operational value for fire management agencies.

4. Expanded geographical coverage: Applying the methodology to the entire Mediterranean region would provide more comprehensive risk assessment and enable cross-regional comparison.

5. Causal analysis: Moving beyond correlation to identify causal relationships between environmental fac-

tors and fire behavior could improve model interpretability and suggest specific interventions for risk reduction.

The methodologies and insights developed in this project contribute to the growing field of machine learning applications for environmental challenges, offering a blueprint for handling similar large-scale prediction tasks in climate science, disaster management, and ecological monitoring.

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